UDP File Downloading System

- Design Report

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1. Design Introduction

This document outlines the implementation of a reliable file transfer system over UDP. The system allows multiple clients to connect to a server, view available files, and download them with guaranteed delivery, even in the presence of network disruptions.

System Architecture

The system is built around a client-server architecture:

1. **Server Component**:
    - TCP listener for accepting client connections and processing commands
    - UDP socket for sending file data
    - Multi-threaded TaskQueue system for handling concurrent clients
    - Session management system for tracking active downloads
    - Monitoring system for detecting and recovering from stalled transfers
2. **Client Component**:
    - TCP connection for sending commands and receiving responses
    - UDP socket for receiving file data
    - Multi-threaded design to separate UI, TCP, and UDP processing
    - File reassembly and tracking system
    - Progress monitoring and reporting

Communication uses:

- **TCP**: For control messages (commands and responses)
- **UDP**: For file data transfer with a reliability layer

Reliable Communication Protocol

Protocol Selection and Implementation

We implemented a **Go-Back-N** protocol for reliable data transfer over UDP, which provides:

- Efficient throughput with sliding window mechanism
- Good balance between complexity and performance
- Automatic recovery from packet loss

Key features of our Go-Back-N implementation:

1. **Sliding Window:**
    - Configurable window size (default: 256 packets, set via WINDOW_SIZE in config.cfg)
    - Sequential packet transmission within the window
    - Window advancement based on received acknowledgments
    - Efficient window boundary management
2. **Sequence Numbering:**
    - Each packet is assigned a sequence number based on file offset
    - Sequence numbers are used for ordering and acknowledgment
    - Enables detection of missing or out-of-order packets
3. **Acknowledgment System:**
    - Client acknowledges each received packet by sequence number
    - Individual per-packet acknowledgment: client sends an ACK for every data packet received, identified by its sequence number
    - Fast acknowledgment processing to maintain throughput
4. **Timeout and Retransmission**:
    - Configurable timeout (default: 250ms, set via ACK_TIMEOUT_MS in config.cfg)
    - Smart retransmission of unacknowledged packets
    - Retry limit to prevent infinite retransmission loops
    - Fixed-timeout retransmission; after MAX_RETRIES (default: 5) are exhausted, the server waits an additional 5-second grace period for any late ACKs before terminating the session
5. **Session Management**:
    - Unique session IDs for each download
    - Comprehensive session state tracking
    - Session monitoring for detecting stalled downloads
    - Graceful session cleanup for completed or abandoned transfers

Message Formats

TCP Messages:

REQ_LISTFILES (Client->Server): 1 byte command (0x04)

Message Type	Command ID	Description	Additional Fields
REQ_LISTFILES	0x4	Request file list	None
RSP_LISTFILES	0x5	Response with file list	Number of files (2 bytes) | List Length (4 bytes) | File entries (variable)
REQ_DOWNLOAD	0x2	Request download	Client IP (4 bytes) | Client Port (2 bytes) | Filename Length (4 bytes) | Filename (variable)
RSP_DOWNLOAD	0x3	Download information	Server IP (4 bytes) | Server Port (2 bytes) | Session ID (4 bytes) | File Length (4 bytes)
REQ_QUIT	0x1	Terminate connection	None
DOWNLOAD_ERROR	0x30	Download error	None

RSP_LISTFILES (Server->Client): 1 byte command (0x05) | 2 bytes number of files | 4 bytes total list length | per file: 4 bytes filename length + N bytes filename

REQ_DOWNLOAD (Client->Server): 1 byte command (0x02) | 4 bytes client IP | 2 bytes client UDP port | 4 bytes filename length | N bytes filename

RSP_DOWNLOAD (Server->Client): 1 byte command (0x03) | 4 bytes server IP | 2 bytes server UDP port | 4 bytes session ID | 4 bytes file size (total: 15 bytes fixed)

DOWNLOAD_ERROR (Server->Client): 1 byte command (0x30) -- sent when the requested file does not exist

REQ_QUIT (Client->Server): 1 byte command (0x01)

UDP Messages:

1. **Data Packet:**
   - Flags (1 byte): LSB 0 indicates data packet
   - Session ID (4 bytes): Identifies the download session
   - File Offset (4 bytes): Position in the file
   - Data Length (4 bytes): Length of the data in this packet
   - File Data (variable): Actual file data
2. **ACK Packet:**
   a. Flags (1 byte): LSB 1 indicates ACK packet
   b. Session ID (4 bytes): Identifies the download session
   c. ACK Number (4 bytes): Sequence number being acknowledged

Transaction Sequence

The typical file download transaction follows this sequence:

1. **Connection Establishment:**
    - Client connects to server via TCP
    - Server accepts connection and creates a client handler thread
2. **File Discovery:**
    - Client sends REQ_LISTFILES command
    - Server processes the request, scans the download directory
    - Server sends RSP_LISTFILES with available files
    - Client displays the file list to the user
3. **Download Initiation:**
    - Client sends REQ_DOWNLOAD with IP, port, and filename
    - Server verifies file existence and prepares for transfer
    - Server assigns a session ID and creates a download session
    - Server sends RSP_DOWNLOAD with server IP, port, session ID, and file size
    - Server spawns a file transfer thread for this session
4. **File Transfer (Go-Back-N Protocol):**
    - Server sends packets within the current window
    - Client receives packets and sends ACKs for each received packet
    - Server advances window based on received ACKs
    - If packets are lost, server retransmits unacknowledged packets after timeout
    - Transfer continues until all file data is sent and acknowledged
    - Client assembles the file from received packets
    - Client reports progress during download

5. **Transfer Completion:**
    - Server confirms all packets have been acknowledged
    - Server closes the download session
    - Client finalizes the downloaded file
    - Client displays download statistics (time, speed)
6. **Error Handling:**
    - If file not found, server sends DOWNLOAD_ERROR
    - If network disruption occurs, retransmission handles recovery
    - If maximum retries are exceeded, server logs error and closes session
    - Client detects and reports stalled downloads

Multi-threading Model

Server-side Threading

1. **Main Thread:**
    - Handles initial setup and TCP listener
    - Accepts incoming client connections
    - Dispatches client connections to the TaskQueue
2. **TaskQueue Worker Threads:**
    - Process TCP commands from clients
    - Handle file listing and download requests
    - Create download sessions and spawn per-session file transfer threads

    Configured with 10 worker threads and a 20-slot bounded queue (matching MAX_CLIENTS and THREAD_POOL_SIZE settings)
3. **File Transfer Threads:**
    - One per active download session
    - Read file data and create packets
    - Handle window management and retransmissions
    - Track acknowledgments and manage session state
4. **UDP Receiver Thread:**
    - Receives and processes ACK packets
    - Updates session state based on acknowledgments

5. **Session Monitor Thread:**
    - Periodically checks for stalled or inactive sessions
    - Cleans up timed-out sessions
    - Logs session statistics

Client-side Threading

1. **Main Thread:**
    - Handles user interface and command input
    - Processes user commands
2. **TCP Receiver Thread:**
    - Receives and processes TCP messages from server
    - Handles file listings and download responses
    - Creates download sessions
3. **UDP Receiver Thread:**
    - Receives and processes UDP data packets
    - Sends acknowledgments
    - Assembles file data
    - Tracks download progress
4. **Session Monitor Thread:**
    - Monitors active downloads for progress
    - Detects stalled downloads
    - Reports download statistics

Optimizations

1. **Socket Buffer Optimization:**
    - Increased socket buffer sizes (1MB) for better performance
    - Reduces the chance of packet loss due to buffer overflow
2. **File I/O Efficiency:**
    - Pre-allocation of file space on the client side
    - Efficient seeking and writing at specific file offsets
    - Buffered I/O for better performance
3. **Protocol Tuning:**
    - Configurable window size for network condition adaptation
    - Adjustable timeout values for different network environments
    - Retry limiting to prevent resource exhaustion

4. **Memory Management:**
    - Reuse of packet buffers to reduce memory allocation overhead
    - Proper buffer sizing based on maximum packet size
    - Efficient vector and map usage for data storage

2. Verification Methodology

2.1 Testing Scenarios

The implementation was verified through comprehensive testing:

1. **Functionality Testing:**
    a. Basic file listing and downloading without network disruption
    b. Multiple simultaneous client connections and downloads
    c. Files of various sizes (small to 200MB)
    d. Different window sizes and protocol configurations
2. **Reliability Testing:**
    a. Network disruption using Ethernet Break Circuit
    b. Multiple brief disconnections during a transfer
    c. Extended disconnection during transfer
    d. High packet loss simulation
    e. Client-side disconnection and reconnection
3. **Performance Testing:**
    a. Large file transfers (100MB+)
    b. Multiple concurrent downloads
    c. Various window sizes to find optimal settings
    d. Different network conditions and latencies
4. **Error Handling Testing:**
    a. File not found scenarios
    b. Invalid commands
    c. Malformed packets
    d. Server shutdown during transfer
    e. Client crash and recovery

2.2 Test Results

Our implementation successfully handled all test scenarios:

1. **Basic Functionality:**
    - File listing worked correctly with directories of various sizes
    - Downloaded files matched source files (verified by checksum)
    - Multiple clients could connect and download simultaneously
2. **Reliability:**
    - Files transferred correctly despite network disruptions
    - Recovery from packet loss was automatic
    - Protocol correctly handled out-of-order packets
3. **Performance:**
    - Achieved good throughput with optimized window sizes
    - Multiple concurrent downloads showed fair resource sharing
    - Large files transferred successfully with consistent speed
4. **Error Handling:**
    - Appropriate error messages displayed for all error conditions
    - No resource leaks observed during stress testing
    - Graceful cleanup of aborted transfers

2.3 Performance Analysis

```
D:\SIT-Digipen\Year 2\Trimest
[20:28:24] Received ACK: Session 3, Seq 6399
[20:28:24] Session 3 progress: 99%
[20:28:24] Sent packet: Session 3, Seq 6400, Offset 104774400, Size 16371
[20:28:24] Sent packet: Session 3, Seq 6401, Offset 104790771, Size 16371
[20:28:24] Sent packet: Session 3, Seq 6402, Offset 104807142, Size 16371
[20:28:24] Sent packet: Session 3, Seq 6403, Offset 104823513, Size 16371
[20:28:24] Sent packet: Session 3, Seq 6404, Offset 104839884, Size 16371
[20:28:24] Sent packet: Session 3, Seq 6405, Offset 104856255, Size 1345
[20:28:24] Received ACK: Session 3, Seq 6400
[20:28:24] Received ACK: Session 3, Seq 6401
[20:28:24] Received ACK: Session 3, Seq 6402
[20:28:24] Received ACK: Session 3, Seq 6403
[20:28:24] Received ACK: Session 3, Seq 6404
[20:28:24] Received ACK: Session 3, Seq 6405
[20:28:24] Timeout for session 3, retry 1/5
[20:28:24] Timeout for session 3, retry 2/5
[20:28:25] Timeout for session 3, retry 3/5
[20:28:25] Timeout for session 3, retry 4/5
[20:28:25] Timeout for session 3, retry 5/5
[20:28:25] Maximum retries limit reached for session 3, waiting 5 more seconds for any final ACKs
[20:28:27] No new ACKs during final wait, ending transfer
[20:28:27] Last chunk (Seq 6405) is ACKed - transfer is complete
[20:28:27] File transfer completed successfully: Session 1
[20:28:28] No new ACKs during final wait, ending transfer
[20:28:28] Last chunk (Seq 6405) is ACKed - transfer is complete
[20:28:28] File transfer completed successfully: Session 2
[20:28:30] No new ACKs during final wait, ending transfer
[20:28:30] Last chunk (Seq 6405) is ACKed - transfer is complete
[20:28:30] File transfer completed successfully: Session 3
```

```
D:\SIT-Digipen\Year 2\Trimest
[20:28:06] Starting download from 192.168.50.193:8002
[20:28:06] Session ID: 1
[20:28:06] File size: 100.00 MB
[20:28:06] Output file: D:\Dest1\download_1
[20:28:06] Session 1: 0% complete, 532.91 KB/s
[20:28:07] Session 1: 5% complete, 7.56 MB/s
[20:28:08] Session 1: 10% complete, 7.81 MB/s
[20:28:08] Session 1: 15% complete, 7.84 MB/s
[20:28:09] Session 1: 20% complete, 7.86 MB/s
[20:28:09] Session 1: 25% complete, 7.75 MB/s
[20:28:10] Session 1: 30% complete, 7.68 MB/s
[20:28:11] Session 1: 35% complete, 7.50 MB/s
[20:28:12] Session 1: 40% complete, 7.32 MB/s
[20:28:12] Session 1: 45% complete, 7.26 MB/s
[20:28:13] Session 1: 50% complete, 7.26 MB/s
[20:28:14] Session 1: 55% complete, 7.20 MB/s
[20:28:15] Session 1: 60% complete, 7.18 MB/s
[20:28:15] Session 1: 65% complete, 7.15 MB/s
[20:28:16] Session 1: 70% complete, 7.14 MB/s
[20:28:17] Session 1: 75% complete, 7.16 MB/s
[20:28:17] Session 1: 80% complete, 7.19 MB/s
[20:28:18] Session 1: 85% complete, 7.20 MB/s
[20:28:19] Session 1: 90% complete, 7.22 MB/s
[20:28:19] Session 1: 95% complete, 7.20 MB/s
[20:28:21] Session 1: 100% complete, 6.90 MB/s
[20:28:21] Download completed: download_1
[20:28:21] Size: 100.00 MB
[20:28:21] Time: 14s
[20:28:21] Speed: 6.90 MB/s
```

```
D:\SIT-Digipen\Year 2\Trimest
[20:28:09] Starting download from 192.168.50.193:8002
[20:28:09] Session ID: 3
[20:28:09] File size: 100.00 MB
[20:28:09] Output file: D:\Dest3\download_3
[20:28:09] Session 3: 0% complete, 207.63 KB/s
[20:28:10] Session 3: 5% complete, 6.63 MB/s
[20:28:11] Session 3: 10% complete, 7.11 MB/s
[20:28:12] Session 3: 15% complete, 6.51 MB/s
[20:28:12] Session 3: 20% complete, 6.64 MB/s
[20:28:13] Session 3: 25% complete, 6.71 MB/s
[20:28:14] Session 3: 30% complete, 6.71 MB/s
[20:28:14] Session 3: 35% complete, 6.74 MB/s
[20:28:15] Session 3: 40% complete, 6.77 MB/s
[20:28:16] Session 3: 45% complete, 6.85 MB/s
[20:28:17] Session 3: 50% complete, 6.85 MB/s
[20:28:17] Session 3: 55% complete, 6.90 MB/s
[20:28:18] Session 3: 60% complete, 6.94 MB/s
[20:28:19] Session 3: 65% complete, 6.99 MB/s
[20:28:19] Session 3: 70% complete, 7.04 MB/s
[20:28:20] Session 3: 75% complete, 6.86 MB/s
[20:28:21] Session 3: 80% complete, 6.70 MB/s
[20:28:22] Session 3: 85% complete, 6.65 MB/s
[20:28:23] Session 3: 90% complete, 6.62 MB/s
[20:28:24] Session 3: 95% complete, 6.66 MB/s
[20:28:24] Session 3: 100% complete, 6.71 MB/s
[20:28:24] Download completed: download_3
[20:28:24] Size: 100.00 MB
[20:28:24] Time: 14s
[20:28:24] Speed: 6.71 MB/s
```

```
D:\SIT-Digipen\Year 2\Trimest
[20:28:07] Starting download from 192.168.50.193:8002
[20:28:07] Session ID: 2
[20:28:07] File size: 100.00 MB
[20:28:07] Output file: D:\Dest2\download_2
[20:28:07] Session 2: 0% complete, 347.55 KB/s
[20:28:08] Session 2: 5% complete, 7.36 MB/s
[20:28:08] Session 2: 10% complete, 7.72 MB/s
[20:28:09] Session 2: 15% complete, 7.84 MB/s
[20:28:10] Session 2: 20% complete, 7.63 MB/s
[20:28:10] Session 2: 25% complete, 7.64 MB/s
[20:28:11] Session 2: 30% complete, 7.11 MB/s
[20:28:12] Session 2: 35% complete, 7.10 MB/s
[20:28:13] Session 2: 40% complete, 7.09 MB/s
[20:28:14] Session 2: 45% complete, 7.05 MB/s
[20:28:14] Session 2: 50% complete, 7.03 MB/s
[20:28:15] Session 2: 55% complete, 7.04 MB/s
[20:28:16] Session 2: 60% complete, 6.99 MB/s
[20:28:16] Session 2: 65% complete, 6.98 MB/s
[20:28:17] Session 2: 70% complete, 7.01 MB/s
[20:28:18] Session 2: 75% complete, 7.05 MB/s
[20:28:18] Session 2: 80% complete, 7.08 MB/s
[20:28:19] Session 2: 85% complete, 7.11 MB/s
[20:28:20] Session 2: 90% complete, 7.04 MB/s
[20:28:21] Session 2: 95% complete, 6.82 MB/s
[20:28:22] Session 2: 100% complete, 6.83 MB/s
[20:28:22] Download completed: download_2
[20:28:22] Size: 100.00 MB
[20:28:22] Time: 14s
[20:28:22] Speed: 6.83 MB/s
```

Performance observations:

- Server can handle multiple download requests simultaneously
- Transfer speed is similar for all clients

3. Individual Contribution

Name	SIT ID	DigiPen ID	Contribution
Bryan Ang Wei Ze	2301397	bryanweize.ang	Responsible for the UDP implementation, Go-Back-N protocol design, file transfer threads, and session management
Tham Kang Ting	2301255	kangting.t	Focused on system architecture, TCP protocol, TaskQueue integration, client-side implementation, and documentation
Low Yue Jun	2301302	yuejun.low	Handled server-side implementation, file management, reliability testing, and error handling